higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

APPLIED ENGINEERING TECHNOLOGY
NQF LEVEL 4

(6021024)

18 November 2019 (X-Paper)
09:00–12:00

Non-programmable scientific calculators may be used.

This question paper consists of 8 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. ALL the sketches must be neat, clear and in good proportion.
5. $g = 9{,}81\ m/s^2$.
6. Write neatly and legibly.

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QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Sir Isaac Newton's third law of motion states:

A when a body keeps its position at rest
B the momentum will increase
C when a man is travelling by car his velocity will increases.
D for every action there is an equal and opposite reaction.

1.1.2 A motion is …

A when an object remains in the same position.
B the distance between two points.
C the change in position of an object with respect to a reference plane.
D when an object increase in velocity

1.1.3 … was/were designed to assess the resistance of a test piece to a wet, salt-filled atmosphere.

A The voltmeter test
B Hydrometer test
C The cylinder test
D The humidity test

1.1.4 The difference between permanent magnets and electromagnets …

A is to supply an electric current through a circuit.
B is to repel each other.
C is that permanent magnets stay magnetic and electromagnets are only magnetic when an electric field is placed near them
D is to attract each other

1.1.5 A … is designed to change the reciprocating motion of the pistons to the rotary motion of the drive wheel.

A rocker shaft
B crankshaft
C input shaft
D camshaft

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1.1.6 Surface finishes are applied to …

A reduce friction.
B increase friction
C generate heat
⌘ D increase expansion

1.1.7 The strength of material is …

A indicated by a number which can change.
B how it combines with other parts.
C to put up barriers and guards.
D the ability of a material to resist an applied force.

1.1.8 Choose the correct value in Pascal of the atmospheric pressure at sea level.

A 104 325
B 103 325
C 102 325
D 101 325 ⌘

1.1.9 The pressure of the atmosphere at the earth's surface is measured with …

A a thermometer.
B a barometer.
C a tachometer
D a micro meter

1.1.10 The main function of a foot valve in a suction pipe is to …

A ensure that fluid does not leave the pump when it is not in operation.
B ensure that fluid leave the pump when it is in operation.
C show the pressure inside the pump.
D stop the driving unit.

(10 × 1) (10)

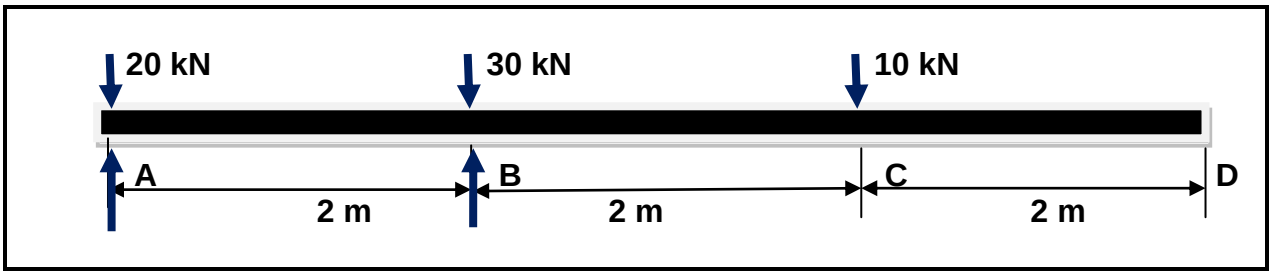
1.2 Name FIVE different types of pumps. (5)

1.3 A body with a mass of 5 tons moves on a horizontal surface. To maintain a constant velocity, a force of 10 kN is used to move the body. ⌘

Calculate the coefficient of friction between the body and the horizontal surface. (5)

[20]

QUESTION 2



2.1 Calculate the reactions at the supports and test your answer. ⌘ (5)

2.2 Use a scale of 1 mm = 1 kN and draw the shear-force diagram. Indicate ALL the values on the diagram. (5)

2.3 Calculate the bending moments at points A, B, C and D. (4)

2.4 Use a scale of 1 mm = 1 kNm and draw the bending moment diagram. Indicate ALL the values on the diagram. ⌘ (5)

2.5 Calculate the position and magnitude of the maximum bending moment. (4)

[23]

QUESTION 3

3.1 Choose a item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–L) next to the question number (3.1.1–3.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	Mechanical device which engage and disengages power transmission especially from driving shaft to driven shaft ⌘	A	trolley jack
		B	clutch
		C	hydraulic fluids
3.1.2	Mechanical device used as a lifting device to lift heavy loads or to apply great forces	D	oil pump
		E	flywheel
3.1.3	Consists of plates or electrodes immersed in a solution or electrolyte	F	piston ⌘
3.1.4	Ability to lubricate moving parts	G	turbine
3.1.5	The clearance or play between two parts such as meshed gears	H	ventilation
		I	backlash
3.1.6	Deliver an adequate volume of oil around the engine	J	battery
3.1.7	A device that converts an electrical signal into mechanical movement	K	reservoir
		L	solenoid
3.1.8	It serves as a mounting surface for the clutch and pressure plate assembly		
3.1.9	It is a moving component that is contained by the cylinder and is made gastight by compression rings ⌘		
3.1.10	Driven fan splined to the input shaft		

⌘ (10 × 1) (10)

3.2 Name FIVE parts of a DC motor. (5)
[15]

QUESTION 4

4.1 Briefly explain the operation of a four-stroke engine during the following strokes:

4.1.1 Inlet or induction stroke

4.1.2 Compression stroke

4.1.3 Power stroke ⌘

4.1.4 Exhaust stroke

(4 × 2) (8)

4.2 A horizontal cylinder with a rod has a diameter of 150 mm. The rod is connected to a piston inside the cylinder. The cylinder is filled with a liquid which has a pressure of 5 kPa ⌘

Calculate the force on the piston when the rod moves forward. (4)

4.3 Safety in the workshop is very important.

⌘ State FIVE safety practices to follow before you operate a drilling machine. (5)

4.4 Name THREE types of electrical motor components. (3)

[20]

QUESTION 5

5.1 State FIVE steps of the design process. (5)

5.2 Define the following terms used in control systems:

5.2.1 Photoelectric smoke detector

5.2.2 Speaker

5.2.3 Motor ⌘

5.2.4 Buzzers

5.2.5 Photodiodes

(5 × 2) (10)

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5.3 Indicate whether the following statements are TRUE or FALSE. Choose the correct answer and write only 'True' or 'False' next to the question number (5.3.1-5.3.7) in the ANSWER BOOK.

5.3.1 The assembly drawing can be used to represent items that consist of more than one component. They show how components fit together. ⌘

5.3.2 Pallets are used to transport components and goods of various types.

5.3.3 Personal protective equipment (PPE) protects your body while you are working. ⌘

5.3.4 A control system consists of components or subsystems that are not connected to each other and work together to achieve a goal.

5.3.5 To create designs that protect your environment, you cannot use biodegradable materials.

5.3.6 Friction is not a force that opposes the motion of two objects moving against each other.

5.3.7 Ensure that all controls on a machine are in a neutral position before you start the machine. ⌘

(7 × 1) (7)
[22]

TOTAL: 100

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